Pergamon

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LENT SOMA SCALES FOR ALL ANATOMIC SITES

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BRAIN

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	
Subjective					
Headache	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Somnolence	Occasional, able to work or perform normal activity	Intermittent, interferes with work or normal activity	Persistent, needs some assistance for self-care	Refractory, prevents daily activity, Coma	____
Intellectual deficit	Minor loss of ability to reason & judge	Moderate loss of ability to reason & judge	Major loss of ability to reason & judge	Complete loss of reasoning and judgment	____
Functional competence	Perform complex tasks with minor inconvenience	Cannot perform complex tasks	Cannot perform simple tasks	Incapable of self-care /supervision, Coma	____
Memory	Decreased short term, difficulty with learning	Decreased long term, loss of short term	Loss of short and long term	Complete disorientation	____
Objective					
Neurologic deficit*	Barely detectable neurologic signs, able to perform normal activities	Easily detectable neurologic abnormalities, interferes with normal activities	Focal motor signs, disturbances in speech, vision, etc., interfering with daily activities	Hemiplegia, hemisensory deficit, aphasia, blindness, etc., requires continuous care, coma	____
Cognitive functions	Minor loss of memory, reason and/or judgment	Moderate loss of memory, reason and judgment	Major intellectual impairment	Complete memory loss and/or incapable of rational thought	____
Mood & personality changes	Occasional & minor	Intermittent & minor	Persistent & minor	Total disintegration	____
Seizures	Focal, without impairment of consciousness	Focal, with impairment of consciousness	Generalized , tonic-clinic or absence attack	Uncontrolled with loss of consciousness > 10 min.	____
Management					
Headache, Somnolence	Occasional non-narcotic medication	Persistent non-narcotic medication, intermittent low dose steroids	Intermittent high dose steroids	Parenteral high dose steroids, mannitol and/or surgery	____
Seizures	Behavioral modification	Behavioral modification and occasional oral medication	Permanent oral medication	Intravenous anti-convulsive medication	____
Cognition, Memory	Minor adaptation	Psychosocial + educational intervention	Occupational and physiotherapy	Custodial care	____

*Function in the irradiated part of brain

SCORING

Instructions

Score the 12 SOM parameters with 1 – 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 12

LENT Score:

_____

BRAIN

Analytic					
Neuropsychologic	Minor deficits in memory, IQ, and/or attention	10 - 19 point decrease in IQ level	20 - 29 point decrease in IQ level	≥ 30 point decease in IQ level, but can learn simple tasks	Y/N Date:
MRI	Focal white matter changes; dystrophic cerebral calcification	White matter changes affecting ≤ 1 cerebral lobe; limited perilesional necrosis	Focal necrosis with mass effect	Pronounced white matter changes; mass effect requiring surgical intervention	Y/N Date:
CT	Assessment of swelling, edema, atrophy				Y/N Date:
MRS	Assessment of chemical spectra				Y/N Date:
PET	Assessment of metabolic activity				Y/N Date:
Magnetic mapping	Assessment of cognitive function				Y/N Date:
Serum	Assessment of myelin basic protein levels				Y/N Date:
CSF	Assessment of total protein and myelin basic protein				Y/N Date:

SPINAL CORD

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Paresthesia (tingling sensation, shooting pain, Lhermitte's syndrome)	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Sensory (numbness)	Minimal change	Mild unilateral sensory loss; works with some difficulties	Partial unilateral sensory loss; needs assistance for self care	Total loss of sensation, danger of self-injury	____
Motor (weakness)	Minor loss of strength	Weakness interfering with normal activities	Persistent weakness preventing basic activities	Paralysis	____
Sphincter control	Occasional loss	Intermittent loss	Incomplete control	Complete incontinence	____
Objective					
Neurologic evaluation	Barely detectable decrease in sensation or motor weakness on one side, no effect on function	Easily detectable decrease in sensation or motor weakness on one side, disturbs but does not prevent function	Full Brown-Sequard syndrome, loss of sphincter function, prevents function	Complete transection, disabling, requiring continuous care	____
Management					
Pain	Occasional non-narcotic medication	Persistent non-narcotic medication, intermittent low dose steroids	Intermittent high dose steroids	Persistent high dose steroids	____
Neurologic function	Needs minor adaptation to continue working	Regular physiotherapy	Intensive physiotherapy plus regular supervision	Intensive nursing and/or life support	____
Incontinence	Occasional use of incontinence pads	Intermittent use of incontinence pads	Regular use of incontinence pads or self-catheterization	Permanent use of pads or catheterization	____

Instructions

Score the 8 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 8

LENT Score:

_____

Analytic						
MRI	Edema	Localized demyelination	Extensive demyelination	Necrosis	Y/N	Date:
CT	Assessment of swelling, edema, atrophy				Y/N	Date:
MRS	Assessment of chemical spectra				Y/N	Date:
PET	Assessment of metabolic activity				Y/N	Date:
Serum	Assessment of myelin basic protein levels				Y/N	Date:
CSF	Assessment of total protein and myelin basic protein				Y/N	Date:

MALE HYPOTHALAMIC / PITUITARY - GONADAL

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective Libido	Occasionally suppressed	Intermittently suppressed	Persistently suppressed	Refractory & excruciating	____ **Instructions** **Score the 4 SOM parameters with 1 - 4**
Objective Fertility Libido	Occasional loss	Intermittent loss	Persistent loss	Impotent	____ ____ **(Score = 0 if there are no toxicities) Total the scores and divide by 4**
Management Libido		Hormone replacement			____ **LENT Score:** _____

Analytic FSH / LH	Normal limits or borderline decreased	Decreased			Y/N Date:
Testosterone	Normal limits or borderline decrease	Decreased			Y/N Date:
Stimulated FSH / LH	Assessment of testes responsiveness and hypothalamic/pituitary - testes axis integrity				Y/N Date: Y/N Date:

FEMALE HYPOTHALAMIC / PITUITARY - GONADAL

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Hot flashes	Occasional	Intermittent	Persistent		____
Dysmenorrhea	Occasional	Intermittent	Persistent		____
Menstruation		Oligomenorrhea	Amenorrhea		____
Libido	Occasionally suppressed	Intermittently suppressed	Persistently suppressed		____
Objective					
Ovulation			Anovulation in premenopausal women		____
Involuntary infertility				Infertile	____
Osteoporosis			Radiographic evidence	Fracture	____
Management					
Dysmenorrhea, Hot flashes		Persistent hormone replacement			____
Menstruation		Hormone replacement			____
Osteoporosis		Hormone replacement, Calcium supplements			____

Instructions

Score the 10 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 10

LENT Score:

_____

Analytic		
FSH / LH/ Estradiol	Assessment of hypothalamic/pituitary - gonadal axis integrity	Y/N Date:
Bone densitometry	Quantify bone density	Y/N Date:
Stimulated FSH / LH	Assessment of pituitary responsiveness	Y/N Date:

HYPOTHALAMIC / PITUITARY - ADRENAL

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Activity level	Occasional fatigue	Intermittent fatigue and drowsiness	Drowsiness and weakness	Paralysis / coma	____
Appetite	Occasional anorexia	Anorexia / nausea	Persistent vomiting	Refractory vomiting	____
Skin color	Darkened scars	Darkened mucosa, palmar creases	Darkened skin		____
Objective					
Strength			Muscle weakness	Paralysis	____
Cardiovascular		BP 20% below baseline	BP 20% - 50% below baseline	BP > 50% below baseline	____
Metabolic	Occasional salt craving and muscle cramping	Intermittent salt craving and muscle cramping, light headedness	Persistent salt craving and muscle cramping, dizziness, syncope	Refractory muscle cramping, coma	____
Skin color	Darkened scars	Darkened mucosa, palmar creases	Darkened skin		____
Management					
Hypoadrenalism	Hydrocortisone replacement				____

Instructions
Score the 8 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the Scores and divide by 8

LENT Score:

_____

Analytic		
Corticotrophin - stimulation test	Assessment of adrenal responsiveness and hypothalamic/pituitary - adrenal axis integrity	Y/N Date:
Corticotrophin - releasing hormone stimulation test	Assessment of adrenal responsiveness and hypothalamic/pituitary - adrenal axis integrity	Y/N Date:

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Vision	Indistinct color vision	Blurred vision, loss of color vision	Severe loss of vision, symptomatic visual field defect with decrease in central vision, some ability to perform daily living activities	Blind, inability to perform daily living activities	____
Light sensitivity	Photophobia, no change in vision	Increased photophobia, decreased vision	Photophobia, major loss of vision		____
Pain / Dryness	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Tearing	Occasional	Intermittent	Persistent		____
Objective					
Best corrected vision	> 20/40	20/50 - 20/200	< 20/200 Count fingers at 1 meter	Cannot count fingers at 1 meter	____
Cornea	Increased tearing on exam	Non-infectious keratitis	Infectious keratitis, corneal ulcer	Panophthalmitis, corneal scar, ulceration leading to perforation/loss of globe,	____
Iris	Rubeosis only	Rubeosis , increased intraocular pressure	Neovascular glaucoma with ability to count fingers at 1 meter	Neovascular glaucoma without ability to count fingers at 1 meter, complete blindness	____
Sclera	Loss of episcleral vessels	≤ 50% scleral thinning	> 50% scleral thinning	Scleral or periosteal graft required due to perforation	____
Optic nerve	Afferent pupillary defect with normal appearing nerve	≤ 1/4 pallor with asymptomatic visual field defect	> 1/4 pallor or central scotoma	Profound optic atrophy, complete blindness	____
Lens	Asymmetric lenticular opacities, no visual loss	Moderate lenticular changes with mild-moderate visual loss	Moderate lenticular changes with severe visual loss	Severe lenticular changes	____
Retina	Microaneurysms, nonfoveal exudates, minor vessel attenuation, extra-foveal pigment changes	Cotton wool spots	Massive macular exudation, focal retinal detachment	Opaque vitreous hemorrhage, complete retinal detachment, blindness	____
Facial bones	Cosmetically undetectable facial asymmetry	Minimal cosmetic asymmetry	Moderate orbital contracture	Severe hypoplasia of orbital bones	____

Instructions
Score the 18 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 18

Management					
Tearing, Cornea, Lacrimation	Lubrication as needed	Lubrication with or without pressure patch. antibiotics	Topical antibiotics with or without cycloplegia	Corneal graft, Enucleation	____
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Parenteral narcotics	____
Neovascularization	Pan-retinal photocoagulation for neovascular changes	Medical management of glaucoma, pan-retinal photocoagulation	Surgical management of glaucoma, Cytodestructive procedure	Enucleation	____
Lens			Cataract extraction depending on visual potential		____
Retina		Medical management of glaucoma, focal photocoagulation	Surgical management of glaucoma, with or without pan-retinal photocoagulation	Cytodestructive procedure, repair of retinal detachment	____
Facial bones			Cosmetic repair ± orbital augmentation for anophthalmic socket	Enucleation, orbital augmentation for anophthalmic socket	____

LENT Score:

_____

Analytic		
Slit lamp exam	Assessment of intraocular pressure, pupils, ocular motility, dilated fundoscopic exam and gonioscopy	Y/N Date:
Cultures and stains	Assessment of corneal infiltrates	Y/N Date:
Ultrasound	Examination of posterior pole if opaque media, i.e., cornea, lens, vitreous	Y/N Date:
Fluorescein angiogram	Evaluation of retinal neovascularization, macular edema/exudates	Y/N Date:
Color vision	Assessment if afferent pupillary defect, or optic nerve asymmetry	Y/N Date:
Automated visual field	Bilateral - assessment of optic nerve, pupillary or color vision abnormality	Y/N Date:
MRI	Assessment of sudden visual loss and abnormal optic disc or normal appearing optic disc and no other visible reason for visual loss	Y/N Date:

EAR

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Tinnitus	Occasional	Intermittent	Persistent	Refractory	____
Hearing	Minor loss, no impairment in daily activities	Frequent difficulties with faint speech	Frequent difficulties with loud speech	Complete deafness	____
Objective					
Skin	Dry desquamation	Otitis externa	Superficial ulceration	Deep ulceration, necrosis, osteochondritis	____
Hearing	< 10 decibel loss in one or more frequencies	10 - 15 decibel loss in one or more frequencies	> 15 - 20 decibel loss in one or more frequencies	> 20 decibel loss in one or more frequencies	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Parenteral narcotics	____
Skin	Occasional lubrication / ointments	Regular eardrops or antibiotics	Eardrums	Surgical intervention	____
Hearing loss			Hearing aid		____

Instructions

Score the 8 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 8

LENT Score:

_____

Analytic		
Pure tone audiometry	Assessment of characteristics of sensorineural perception.	Y/N Date:
Speech audiometry	Assessment of characteristics of speech perception	Y/N Date:

MUCOSA - ORAL AND PHARYNGEAL

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Dysphagia	Difficulty eating solid food	Difficulty eating soft food	Can take liquids only	Totally unable to swallow	____
Taste alteration	Occasional, slight	Intermittent	Persistent		____
Objective					
Mucosal integrity	Patchy atrophy or telangiectasia	Diffuse atrophy or telangiectasia, superficial ulcer	Deep ulcer no bone or cartilage exposure	Deep ulcer with bone or cartilage exposure	____
Weight	≤ 5% loss	>5% - 10% loss	> 10% - 15% loss	> 15% loss	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Ulcer		Cleanse	Antibiotics or oxidants	Debridement + other surgical intervention	____
Dysphagia	Lubricants, diet modification	Non-narcotic	Narcotic	PEG tube and/or surgical intervention	____
Taste alteration	Minor diet changes (non-acidic)	Minor diet changes (semi-soft)	Major diet changes (soft)	Major diet changes (liquid)	____

Instructions
Score the 9 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 9

LENT Score:

_____

Analytic		
Color photo	Assessment of changes in appearance	Y/N Date:
Cytology, Biopsy, Imaging	Rule out persistent tumor	Y/N Date:
Smear, Culture, Antifungal trial	Rule out candidiasis	Y/N Date:

SALIVARY GLAND

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective Xerostomia	Occasional dryness	Partial but persistent dryness	Complete dryness, non-debilitating	Complete dryness, debilitating	____ **Instructions** **Score the 3 SOM parameters with 1 - 4**
Objective Saliva	Normal moisture	Scant saliva	Absence of moisture, sticky, viscous saliva	Absence of moisture, coated mucosa	____ **(Score = 0 if there are no toxicities)** **Total the scores and divide by 3**
Management Xerostomia		Occasional saliva substitute Sugarless candy or gum, Sialogogues	Frequent saliva substitute or water Sugarless candy or gum, Sialogogues	Needs saliva substitute or water in order to eat Sugarless candy or gum, Sialogogues	____ **LENT Score:** _____

Analytic Salivary flow/quantity/ stimulation	76% - 95% of pre-treatment	51% - 75% of pre-treatment	26% - 50% of pre-treatment	0% - 25% of pre-treatment	Y/N Date:

MANDIBLE

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Mastication		Difficulty with solids	Difficulty with soft foods		____
Denture use		Loose dentures	Inability to use dentures		____
Trismus	Noted but unmeasurable	Preventing normal eating	Difficulty eating	Inadequate oral intake	____
Objective					
Exposed bone		≤ 2 cm	> 2 cm or limited sequestration	Fracture	____
Trismus		1 cm - 2 cm opening	0.5 cm - 1 cm opening	< 0.5 cm opening	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention or resection	____
Exposed bone		Antibiotics	Debridement, HBO_2	Resection	____
Trismus & mastication		Soft diet	Liquid diet, Antibiotics, Muscle relaxant meds	NG tube, gastrostomy	____

Instructions

Score the 9 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 9

LENT Score:

_____

Analytic Mandibular radiograph	Questionable changes or none	Osteoporosis (radiolucent) Osteosclerosis (radiodense)	Sequestra	Fracture	Y/N Date:
Panograph X-rays / CT	Assessment of necrosis progression				Y/N Date:

TEETH

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____ **Instructions** **Score the 4 SOM parameters with 1 - 4**
Objective					
Caries	DMF* < 25%	DMF* 25% - 50%	DMF* > 50%	Fracture	____ **(Score = 0 if there are no toxicities)**
					Total the scores and divide by 4
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Extraction	____ **LENT Score:**
Caries	Prescribed fluorides	Restoration	Selected extraction	Total extraction	____ _____

*DMF = # of decayed, missing, filled teeth compared to pre-RT

Analytic		
Dental x-rays	Assessment of necrosis progression with periapical, bite wing, panograph x-rays	Y/N Date:
Probe for softness	Assessment of teeth for decalcification and caries	Y/N Date:
Pulp test	Assessment of pulp integrity with heat, cold, electric current	Y/N Date:
Percussion	Assessment for tenderness and pain	Y/N Date:
Mobility	Assessment of alveolar bone for loss and infection	Y/N Date:

LARYNX

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Voice / Hoarseness	Occasional hoarseness on prolonged use	Intermittent hoarseness, voice unreliable, varies in day-to-day communication	Persistent hoarseness, incapable of normal communication	Complete loss of voice	____
Breathing	Occasional difficulty	Intermittent difficulty	Labored breathing	Stridor	____
Objective					
Edema	Arytenoids only	Arytenoids and aryepiglottic folds	Diffuse edema of supraglottis, airway adequate	Diffuse with significant narrowing of airway, <1/2 normal	____
Mucosal integrity	Patchy atrophy, telangiectasia	Complete atrophy, extensive telangiectasia	Ulcer, cartilage not exposed	Necrosis, cartilage exposed	____
Respiration		Dyspnea on exertion	Labored at rest	Stridor at rest	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Hoarseness		Rest voice, or whisper only	No talking or whispering	Laryngectomy	____
Respiration		Humidifier, steroids	Temporary tracheostomy	Permanent tracheostomy	____

Instructions
Score the 9 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 9

LENT Score:

_____

Analytic		
Indirect laryngoscopy	Assessment of edema, mucosal integrity, vocal cord motion, ulcer, and necrosis	Y/N Date:
Direct laryngoscopy	Assessment of edema, mucosal integrity, vocal cord motion, ulcer, and necrosis	Y/N Date:
CT	Assessment of edema, necrosis, asymmetry	Y/N Date:
MRI	Assessment of edema, necrosis, asymmetry	Y/N Date:

THYROID & HYPOTHALAMIC / PITUITARY - THYROID AXIS

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Metabolic	Occasional chilliness**	Intermittent chilliness	Needs supplemental heat		____
Gastrointestinal	Occasional constipation**	Intermittent constipation	Persistent constipation		____
Weight	≥ 5% gain**	< 10 % gain	≥ 10% gain		____
Skin texture		Intermittent sensation of dryness	Persistent sensation of dryness		____
Energy level	Occasional fatigue**	Intermittent fatigue	Persistent fatigue		____
Objective					
Facies		Barely noticeable puffiness and thickened lips	Obvious puffiness and thickened lips		____
Speech quality		Barely noticeable hoarseness and slowed speech	Obvious hoarseness and slowed speech		____
Skin temperature		Cool	Cold		____
Hair texture		Difficult to comb	Brittle, splitting, hair loss		____
Nodules				Palpable	____
Heart rate			Slowed		____
Management					
All SOM symptoms		Thyroid replacement therapy			____
Nodules			Surgery / radionuclide therapy		____

Instructions

Score the 13 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 13

LENT Score:

_____

* Primary thyroid vs ** Hypothalamic / Pituitary - Thyroid Axis

Analytic					
Basal T4	Normal limits	0% - 50% decrease	> 50% decrease		Y/N Date:
Basal TSH*	Increased				Y/N Date:
Basal TSH**	Decreased				
Stimulated TSH*	Assessment of thyroid responsiveness				Y/N Date:
Stimulated TSH**	Assessment of pituitary responsiveness and hypothalamic/pituitary-thyroid axis integrity				

BREAST

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal Hypersensation, Pruritus	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Objective					
Edema	Asymptomatic	Symptomatic	Secondary dysfunction		____
Fibrosis* / Fat necrosis	Barely palpable increased density	Definite increased density and firmness	Very marked density, retraction and fixation		____
Telangiectasia	< 1 cm^2	1 cm^2 - 4 cm^2	> 4 cm^2		____
Lymphedema, arm (circumference)	2 cm - 4 cm increase	> 4 cm - 6 cm increase	> 6 cm increase	Useless arm, angiosarcoma	____
Retraction/Atrophy**	10% - 25%	> 25% - 40 %	> 40% - 75%	Whole breast	____
Ulcer	Epidermal only, ≤ 1 cm^2	Dermal, > 1 cm^2	Subcutaneous	Bone exposed, necrosis	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Edema			Medical intervention	Surgical intervention/ mastectomy	____
Lymphedema, arm		Elevate arm, elastic stocking	Compression wrapping, intensive physiotherapy	Surgical intervention/ amputation	____
Atrophy				Surgical intervention/ mastectomy	____
Ulcer		Medical intervention	Surgical intervention, wound debridement	Surgical intervention/ mastectomy	____

Instructions

Score the 12 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 12

LENT Score:

_____

* Compare exposed area to contralateral non-irradiated skin according to defined parameters ** Volume loss due to surgery +/or RT. (compared to opposite breast)

Analytic		
Photographs	Assessment of skin changes as atrophy, retraction or fibrosis, ulcer	Y/N Date:
Tape measure	Assessment of breast size and forearm diameter	Y/N Date:
Mammogram	Assessment of skin thickness and breast density	Y/N Date:
CT/MRI	Assessment of breast size, fat atrophy, and fibrosis density	Y/N Date:

HEART

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Angina pectoris	Occasional, only with intense exertion	With moderate exertion	With mild exertion	At rest	____
Pericardial Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Palpitation	Occasional	Intermittent	Persistent	Refractory	____
Dyspnea	SOB on intense exertion	SOB on mild exertion	SOB at rest, limits all activity	Prevents any physical activity	____
Pedal edema		Asymptomatic	Symptomatic	Prevents daily activities	____
Objective					
Pedal edema	1+	2+	3+	4+	____
Cardiomegaly	Minimal enlargement of cardiosilhouette (ECS)	ECS without pulmonary congestion	ECS with minimal pulmonary congestion	ECS with frank pulmonary edema	____
Cardiac dysrhythmia	Occasional, asymptomatic	Intermittent ECG changes	Persistent ECG changes	Refractory	____
Myocardial CHF	Asymptomatic decline of resting ejection fraction by ≤20% of baseline	Decline of resting ejection fraction by >20% of baseline	Reversible CHF	Irreversible CHF	____
Myocardial ischemia	Abnormal stress test NL resting EKG	Asymptomatic, ST & T wave changes without stress test	Angina without evidence for infarction	Acute myocardial infarction	____
Pericardial disease	Asymptomatic effusion	Rub, chest pain, ECG changes	Tamponade	Constriction	____
Management					
Pain (pericarditis)	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Coronary artery bypass	____
Angina	Present but no therapy	Nitroglycerin PRN	Long acting agents	Coronary artery bypass	____
Pericardial disease		Present but no therapy	Pericardiocenthesis	Pericardiectomy	____
Cardiac dysrhythmia			Medical intervention	Requires monitoring or cardioversion	____
Myocardial infarction			Medical intervention	Coronary bypass	____
Myocardial CHF			Medical intervention	Cardiac transplant	____

Instructions

Score the 17 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 17

LENT Score:

_____

HEART

Analytic Equilibrium radionuclide Angiography (ERNA)	Abnormal, < 20% decrease in left ventricle ejection fraction	20% - 40% decrease in left ventricle ejection fraction	< 40% decrease in left ventricle ejection fraction		Y/N Date:
Exercise Tolerance Test (ETT)	Assessment of pulse rate, blood pressure and EKG polarization changes				Y/N Date:
Cardiac catheterization	Assessment of coronary artery blood flow				Y/N Date:
Thallium Scintigraphy	Assessment of myocardial perfusion				Y/N Date:
Coronary angiography	Assessment of number of vessels involved and extent of stenosis				

VESSELS

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Arterial	No clinical symptom	Clinical symptoms of ischemia during exertion	Clinical symptoms of ischemia at rest	Necrosis	____
Venous	Asymptomatic	Thrombosis requiring no systemic anticoagulant therapy	Thrombosis requiring systemic anticoagulant therapy	Pulmonary embolism and/or thrombosis requiring surgical intervention	____
Objective					
Arterial	Minor ischemia	Intermittent ischemia	Intense ischemia	Necrosis	____
Venous	Minor edema	Intermittent edema	Ulcus or active thrombosis	Pulmonary embolism	____
Management					
Arterial	Behavioral changes	Permanent medicam treatment	Conservative surgery	Amputation	____
Venous	Elastic bandage	Elastic bandage and regular antibiotics / anti-coagulant treatments	Regular parenteral anticoagulant treatment	Surgery	____

Instructions

Score the 6 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 6

LENT Score:

_____

Analytic		
Doppler Ultrasound	Assessment of blood flow and detection of abnormalities	Y/N Date:
Angiography	Assessment of lumen size and collateralization	Y/N Date:

LUNG

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Cough	Occasional	Intermittent	Persistent	Refractory	____
Dyspnea	Breathless on intense exertion	Breathless on mild exertion	Breathless at rest, limits all activities	Prevents any physical activity	____
Chest pain/discomfort	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Objective					
Pulmonary fibrosis	Radiological abnormality	Patchy dense abnormalities on radiograph	Dense confluent radiographic changes limited to radiation field	Dense fibrosis, severe scarring & major retraction of normal lung	____
Lung function	10% - 25% reduction of respiration volume and/or diffusion capacity	> 25% - 50% reduction of respiration volume and/or diffusion capacity	> 50% - 75% reduction of respiration volume and/or diffusion capacity	> 75% reduction of respiration volume and/or diffusion capacity	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Cough		Non-narcotic	Narcotic, intermittent corticosteroids	Respirator, continuous corticosteroids	____
Dyspnea		Occasional O_2	Continuous O_2		____

Instructions

Score the 8 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 8

LENT Score:

_____

Analytic					
PFT	Decrease to >75% - 90% of preTx value	Decrease to >50% - 75% of preTx value	Decrease to >25% - 50% of preTx value	Decrease to ≤ 25% of preTx value	Y/N Date:
DLCO	Decrease to >75% - 90% of preTx value	Decrease to >50% - 75% of preTx value	Decrease to >25% - 50% of preTx value	Decrease to ≤ 25% of preTx value	Y/N Date:
% O_2/CO_2 saturation	> 70% O_2, ≤ 50% CO_2	> 60% O_2, ≤ 60% CO_2	> 50% O_2, ≤ 70% CO_2	≤50% O_2, >70% CO_2	Y/N Date:
CT/ MRI	Assessment of lung volume and zones of fibrosis				Y/N Date:
Perfusion scan	Assessment of pulmonary blood flow and alveolar filling				Y/N Date:
Lung lavage	Assessment of cells and cytokines				Y/N Date:

6901

ESOPHAGUS

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					**Instructions**
Dysphagia	Difficulty eating solid foods	Difficulty eating soft foods	Can take liquids only	Totally unable to swallow	____ **Score the 11 SOM parameters with 1 - 4**
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Objective					
Weight loss from time of treatment	≥ 5% - 10%	> 10% - 20%	> 20% - 30%	> 30%	____ **(Score = 0 if there are no toxicities)**
Stricture	> 2/3 normal diameter with dilatation	> 1/3 - 2/3 normal diameter with dilatation	≤ 1/3 normal diameter	Complete obstruction	____
Ulceration	Superficial ≤ 1 cm^2	Superficial > 1 cm^2	Deep ulcer	Perforation, fistulae	____ **Total the scores and divide by 11**
Bleeding (melena or hematemesis)	Occult	Occasional, normal hemoglobin	Intermittent, 10% - 20% decrease in hemoglobin	Persistent, > 20% decrease in hemoglobin	____
Anemia		Fatigue	Exhaustion		____
Management					
Dysphagia / Stricture	Diet modification or antacids	Diet modification and occasional dilatation	Temporary NG tube or regular dilatation	Parenteral feeding, prosthesis, gastrostomy or permanent NG tube	____ **LENT Score:**
Weight loss	Diet modification	Nutritional supplements	Tube feeding	Surgical bypass, PEG	____ _____
Pain / Ulceration	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Bleeding	Iron therapy	Occasional transfusion	Frequent transfusions	Surgical intervention	____

Analytic		
Barium esophagram	Assessment of esophageal lumen, stricture, dilatation	Y/N Date:
Endoscopy	Assessment of esophageal lumen, mucosal integrity, ulceration	Y/N Date:
CT	Assessment of esophageal wall thickness, lumen, stricture, dilatation	Y/N Date:
MRI	Assessment of esophageal wall thickness, lumen, stricture, dilatation	Y/N Date:
Ultrasonography	Assessment of esophageal wall thickness, lumen, stricture, dilatation	Y/N Date:
Mobility esophagram	Assessment of motility of bolus and peristalsis	Y/N Date:
Electromyogram	Assessment of motility of bolus and peristalsis	Y/N Date:

STOMACH

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Epigastric distress	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Emesis	Occasional	Intermittent	Persistent	Refractory	____
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Objective					
Hematemesis	Occasional	Intermittent	Persistent	Refractory	____
Weight loss from time of treatment	≥ 5% - 10%	> 10% - 20%	> 20% - 30%	> 30%	____
Melena	Occult / Occasional, normal hemoglobin	Intermittent, < 10% decrease in hemoglobin	Persistent, 10% - 20% decrease in hemoglobin	Refractory or frank blood, > 20% decrease in hemoglobin	____
Ulceration	Superficial, ≤ 1 cm^2	Superficial, > 1 cm^2	Deep ulcer	Perforation, fistulae	____
Stricture (Antro-pyloric region)	> 2/3 normal diameter	1/3 - 2/3 normal diameter	< 1/3 normal diameter	Complete obstruction	____
Management					
Epigastric distress / Emesis	Diet modification, antacids	Intermittent prescription medication	Persistent medical management	Surgical intervention	____
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Bleeding	Iron therapy	Occasional transfusion	Frequent transfusions	Embolization, coagulation or surgical intervention	____
Ulceration			Medical intervention	Surgical intervention	____
Stricture			Medical intervention	Surgical intervention	____

Instructions

Score the 13 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 13

LENT Score:

_____

Analytic		
Barium radiography	Assessment of lumen and peristalsis	Y/N Date:
Endoscopy	Assessment of lumen and mucosal surface	Y/N Date:
CT	Assessment of wall thickness, sinus and fistula formation	Y/N Date:
MRI	Assessment of wall thickness, sinus and fistula formation	Y/N Date:

SMALL INTESTINE / COLON

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Stool frequency	2 - 4 per day	5 - 8 per day	> 8 per day	Refractory diarrhea	____
Stool consistency	Bulky	Loose	Mucous, dark, watery		____
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory / Rebound	____
Constipation	3 - 4 per week	Only 2 per week	Only 1 per week	No stool in 10 days	____
Objective					
Melena	Occult / Occasional	Intermittent & tolerable, normal hemoglobin	Persistent, 10% - 20% decrease in hemoglobin	Refractory or frank blood, >20% decrease in hemoglobin	____
Weight loss from time of treatment	≥ 5% - 10%	> 10% - 20%	> 20% - 30%	> 30%	____
Stricture	> 2/3 normal diameter with dilatation	1/3 - 2/3 normal diameter with dilatation	< 1/3 normal diameter	Complete obstruction	____
Ulceration	Superficial ≤ 1 cm^2	Superficial > 1 cm^2	Deep ulcer	Perforation, fistulae	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Stool consistency / frequency	Diet modification	Regular use of non-narcotic antidiarrheal	Continuous use of narcotic antidiarrheal		____
Bleeding	Iron therapy	Occasional transfusion	Frequent transfusions	Surgical intervention	____
Stricture	Occasional diet adaptation	Diet adaptation required	Medical intervention, NG suction	Surgical intervention	____
Ulceration			Medical intervention	Surgical intervention	____

Instructions

Score the 13 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 13

LENT Score:

_____

Analytic		
CT	Assessment of wall thickness, sinus and fistula formation	Y/N Date:
MRI	Assessment of wall thickness, sinus and fistula formation	Y/N Date:
Absorption studies	Assessment of protein and fat absorption and metabolic balance	Y/N Date:
Barium radiograph	Assessment of lumen and peristalsis	Y/N Date:

RECTUM

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Tenesmus	Occasional urgency	Intermittent urgency	Persistent urgency	Refractory	____
Mucosal loss,	Occasional	Intermittent	Persistent	Refractory	____
Sphincter control	Occasional	Intermittent	Persistent	Refractory	____
Stool frequency	2 - 4 per day	4 - 8 per day	> 8 per day	Uncontrolled diarrhea	____
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Objective					
Bleeding	Occult	Occasionally > 2/week	Persistent/daily	Gross hemorrhage	____
Ulceration	Superficial ≤ 1 cm^2	Superficial > 1 cm^2	Deep ulcer	Perforation, Fistulae	____
Stricture	> 2/3 normal diameter with dilatation	1/3 - 2/3 normal diameter with dilatation	< 1/3 normal diameter	Complete obstruction	____
Management					
Tenesmus & stool frequency	Occasional, ≤ 2 antidiarrheals/week	Regular, > 2 antidiarrheals/week	Multiple, > 2 antidiarrheals/day	Surgical intervention/ Permanent colostomy	____
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Bleeding	Stool softener, iron therapy	Occasional transfusion	Frequent transfusions	Surgical intervention / Permanent colostomy	____
Ulceration	Diet modification, stool softener	Occasional steroids	Steroids per enema, hyperbaric oxygen	Surgical intervention/ Permanent colostomy	____
Stricture	Diet modification	Occasional dilatation	Regular dilatation	Surgical intervention/ Permanent colostomy	____
Sphincter control	Occasional use of incontinence pads	Intermittent use of incontinence pads	Persistent use of incontinence pads	Surgical intervention/ Permanent colostomy	____

Instructions
Score the 14 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 11

LENT Score: _____

Analytic		
Barium enema	Assessment of lumen and peristalsis	Y/N Date:
Proctoscopy	Assessment of lumen and mucosal surface	Y/N Date:
CT	Assessment of wall thickness, sinus and fistula formation	Y/N Date:
MRI	Assessment of wall thickness, sinus and fistula formation	Y/N Date:
Anal manometry	Assessment rectal compliance	Y/N Date:
Ultrasound	Assessment of wall thickness, sinus and fistula formation	Y/N Date:

LIVER

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain RUQ	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Objective					
Abdominal findings	Hepatomegaly	Soft ascites	Tense ascites		____
Edema	Occasional leg edema	Intermittent leg edema	Anasarca responsive to diuretics	Anasarca unresponsive to diuretics	____
Weight gain		≤ 5%	> 5% - 10%	> 10%	____
Alertness		Change in attentiveness and sleep pattern	Confusion	Coma	____
Bleeding			Correctable	Unresponsive	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Continuous narcotic	____
Abdominal findings		Intermittent diuretics	Permanent diuretics		____
Bleeding		Iron therapy	Occasional transfusion of fresh frozen plasma	Frequent transfusions	____

Instructions

Score the 9 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 9

LENT Score

_____

Analytic						
AST/ALT /Alk phos	< 2.5 x normal	2.5 - 5.0 x normal	> 5.0 - 20.0 x normal	> 20.0 x normal	Y/N	Date:
Bilirubin	< 1.5 x normal	1.5 - 5.0 x normal	> 5.0 - 10.0 x normal	> 10.0 x normal	Y/N	Date:
PT / PTT	< 1.25 x normal	1.25 - 1.5 x normal	> 1.50 - 2.0 x normal	> 2.0 x normal	Y/N	Date:
Serum alb (gm/dl)[3]	> 3.0	> 2.5 - 3.0	> 2.0 - 2.5	≤ 2.0	Y/N	Date:
Platelets (1,000)	> 75.0	> 50.0 - 75.0	> 25.0 - 50.0	≤ 25.0	Y/N	Date:

KIDNEY

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Symptoms			Fatigue, headache	Obtundation, oliguria, edema	____
Objective					
Blood pressure		systolic ≤ 20 over normal diastolic ≤ 10 over normal	systolic > 20 over normal diastolic > 10 over normal	Malignant hypertension	____
Hematuria	Microscopic	Intermittent macroscopic	Persistent macroscopic	Refractory	____
Edema	None or transient	Pedal; 2+ - 3+	Pedal 4+, edema of lower leg(s)	Uremic coma, anasarca	____
Specific gravity		Urine SpG decreased			____
Management					
Blood pressure/ Renal failure	Diet	Antihypertension medication	Dialysis, unilateral nephrectomy	Permanent dialysis or Renal transplant	____
Hematuria	Iron therapy	Occasional transfusion or single cauterization	Persistent transfusion or coagulation	Surgical intervention	____

Instructions

Score the 7 SOM parameters with 1 - 4

Score = 0 if there are no toxicities)

Total the scores and divide by 7

LENT Score:

_____

Analytic					
Proteinuria	< 3 gm/l	3 gm/l - 10 gm/l	> 10 gm/l	Nephrotic syndrome	Y/N Date:
Creatinine clearance	5% - 10% decrease	> 10% - 30% decrease	> 30% - 60% decrease	> 60% decrease	Y/N Date:
Creatinine	1.25 x normal - 2.5 x normal	> 2.5 x normal - 5 x normal	> 5 x normal - 10 x normal	> 10 x normal	Y/N Date:
B2 Microglob.			> 2 x normal - 4 x normal	> 4 x normal	Y/N Date:
Glomerular filtration rate	Quantification of filtration rate				Y/N Date:
Renal scanning	Assessment of renal size and radioisotope clearance				Y/N Date:

URETER

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____ **Instructions** **Score the 5 SOM parameters with 1 - 4**
Objective Obstruction	Ureteral narrowing without hydronephrosis	Ureteral narrowing with hydronephrosis	Unilateral obstruction	Bilateral obstruction	____ **(Score = 0 if there are no toxicities)**
Renal function	1+ proteinuria	2+ proteinuria	4+ proteinuria		____ **Total the scores and divide by 5**
Management Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____ **LENT Score:**
Obstruction			Unilateral stent or nephrostomy	Bilateral nephrostomy or diversion	____ _____

Analytic Intravenous pyelography	Assessment of ureter integrity	Y/N Date:

BLADDER / URETHRA

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Dysuria	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Frequency	3 - 4 hour intervals	2 - 3 hour intervals	1 - 2 hour intervals	Hourly	____
Hematuria	Occasional	Intermittent	Persistent with clot	Refractory	____
Incontinence	< weekly episodes	< daily episodes	≤2pads/undergarments/day	Refractory	____
Decreased stream	Occasionally weak	Intermittent	Persistent but incomplete obstruction	Complete obstruction	____
Objective					
Hematuria	Microscopic, normal hemoglobin	Intermittent macroscopic, < 10% decrease in hemoglobin	Persistent macroscopic, 10% - 20% decrease in hemoglobin	Refractory, > 20% decrease in hemoglobin	____
Endoscopy	Patchy atrophy or Telangiectasia without bleeding	Confluent atrophy or Telangiectasia with gross bleeding	Ulcerations into muscle	Perforation, fistula	____
Maximum volume	> 300 cc - 400 cc	> 200 cc - 300 cc	> 100 cc - 200 cc	≤ 100 cc	____
Residual volume	25 cc	> 25 cc - 100 cc	> 100 cc		____
Management					
Dysuria	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Frequency	Alkalization	Occasional anti-spasmodic	Regular narcotic	Cystectomy	____
Hematuria/ Telangiectasia	Iron therapy	Occasional transfusion or single cauterization	Frequent transfusion or coagulation	Surgical intervention	____
Incontinence	Occasional use of incontinence pads	Intermittent use of incontinence pads	Regular use of pad or self-catheterization	Permanent catheter	____
Decreased stream		< Once-a-day self-catheterization	Dilatation, > once-a-day self-catheterization	Permanent catheter, surgical intervention	____

Instructions

Score the 14 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 14

LENT Score:

_____

Analytic		
Cystography	Assessment of mucosal surface	Y/N Date:
Volumetric analysis	Assessment of bladder capacity in milliliters	Y/N Date:
Contrast radiography	Assessment for ulcers, capacity and contractility	Y/N Date:
Ultrasound	Assessment of wall thickness, sinus and fistula formation	Y/N Date:
Electromyography	Assessment of sphincter activity using intralumenal pressure transducer, contraction pressure and volume curves	Y/N Date:

TESTES

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective Libido	Occasionally suppressed	Intermittently suppressed	Persistently suppressed		____
Objective Fertility			Oligospermia	Azoospermia	____
Appearance				Atrophy	____
Management Fertility			In vitro fertilization	Sperm retrieval if previously banked	____
Libido		Testosterone			____

Instructions

Score the 5 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 5

LENT Score:

_____

Analytic FSH / LH	Increased FSH / nl LH	Increased FSH / increased LH			Y/N Date:
Testosterone				Decreased	Y/N Date:
Spermatoanalysis	Assessment of sperm number, motility and morphology				Y/N Date:
Flow cytometric analysis	Assessment of sperm chromatin integrity				Y/N Date:
DNA chromatin analysis	Assessment of sperm chromatin integrity				Y/N Date:

SEXUAL DYSFUNCTION - MALE

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Erectile function for vaginal penetration	Occasionally insufficient	Intermittently insufficient	Not sufficient	Impotent	____
Dryness	Occasional	Intermittent	Persistent	Refractory	____
Desire	Occasional	Intermittent	Seldom	Never	____
Satisfaction	Occasional	Intermittent	Seldom	Never	____
Objective					
Frequency		Decreased from normal	Rare	Never	____
Orgasm	Occasional	Intermittent	Seldom	Never	____
Management					
Impotence		Medical intervention	Surgical intervention		____

Instructions

Score the 7 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 7

LENT Score:

_____

Analytic Psychosocial	Evaluate Quality of Life / Sexual Satisfaction	Y/N Date:

VULVA

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Dryness	Occasional	Intermittent	Persistent		____
Pruritus	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Objective					
Pigmentation change	Patchy	Confluent			____
Alopecia	Partial	Complete			____
Atrophy	Patchy	Confluent			____
Appearance	Telangiectasia without bleeding	Telangiectasia with gross bleeding			____
Ulceration / necrosis	Superficial ≤ 1 cm^2	Superficial > 1 cm^2	Deep	Fistulae	____
Fibrosis			Partial	Complete	____
Edema			Partial	Complete	____
Introital stenosis			Partial	Complete	____
Serous transudate	Occasional	Intermittent	Persistent	Refractory	____
Management					
Pruritus / Atrophy	Occasional hormone cream	Intermittent hormone cream	Regular hormone cream		____
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Ulceration	Conservative	Wound care	Debridement	Graft	____
Introital stenosis	Occasional dilation	Intermittent dilation	Persistent Dilation	Surgical repair	____

Instructions

Score the 16 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 16

LENT Score:

_____

Analytic Color photograph	Assessment of changes in skin, mucous and telangiectasia	Y/N Date:

VAGINA

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					**Instructions**
Dyspareunia	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____ **Score the 17 SOM parameters with 1 - 4**
Dryness	Occasional	Intermittent	Persistent	Refractory	____
Bleeding	Occasional	Intermittent	Persistent	Refractory	____
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Objective					
Stenosis / length	> 2/3 normal length	1/3 - 2/3 normal length	< 1/3 normal length	Obliteration	____ **(Score = 0 if there are no toxicities)**
Dryness	Asymptomatic	Symptomatic	Secondary dysfunction		____
Ulceration / necrosis	Superficial, ≤ 1 cm^2	Superficial, > 1 cm^2	Deep ulcer	Fistulae	____
Atrophy	Patchy	Confluent	Nonconfluent	Diffuse	____
Appearance	Telangiectasia without bleeding	Telangiectasia with gross bleeding			____
Synechiae		Partial	Complete		____ **Total the scores and divide by 17**
Bleeding		On contact	Intermittent	Persistent	____
Management					
Dyspareunia/ Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____ **LENT Score:**
Atrophy	Occasional hormone cream	Intermittent hormone cream	Regular hormone cream		____ _____
Bleeding	Iron therapy	Occasional transfusion	Frequent transfusions	Surgical intervention	____
Stenosis	Occasional dilation	Intermittent dilation	Persistent dilation	Surgical reconstruction	____
Dryness	Hormone replacement	Artificial lubrication			____
Ulceration	Conservative	Debridement	HBO_2	Graft, Surgical repair	____
Analytic					
MRI	Assessment of wall thickness, sinus and fistula formation				Y/N Date:
Ultrasound	Assessment of wall thickness, sinus and fistula formation				Y/N Date:
EUA Cytology / biopsy	Assessment of wall diameter and length and mucosal surface				Y/N Date:

UTERUS / CERVIX

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					**Instructions**
Amenorrhea	Asymptomatic	Symptomatic		Infertility	____ **Score the 18 SOM parameters with 1 - 4**
Dysmenorrhea	Asymptomatic	Symptomatic			____
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Bleeding	Occasional, normal hemoglobin	Intermittent < 10% decrease in hemoglobin	Persistent, 10% - 20% decrease in hemoglobin	Refractory > 20% decrease in hemoglobin	____
Objective					
Pyometra	Asymptomatic	Symptomatic			____ **(Score = 0 if there are no toxicities)**
Hematometra	Asymptomatic	Symptomatic			____
Necrosis	Asymptomatic	Symptomatic			____
Ulceration	Superficial, ≤ 1 cm^2	Superficial, > 1 cm^2	Deep ulcer	Fistulae	____ **Total the scores and divide by 18**
Incompetent				Infertility	____
Cervical Os Stenosis	Asymptomatic	Symptomatic			____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____ **LENT Score**
Amenorrhea, Dysmenorrhea, Hematometra	Occasional hormone replacement	Intermittent hormone replacement	Persistent hormone replacement		____
Pyometra		D&C, antibiotics			____
Necrosis		Debridement	D&C	Hysterectomy	____ _____
Bleeding	Iron therapy	Occasional transfusion	Frequent transfusions	Surgical intervention	____
Cervical Os Stenosis			D&C	Hysterectomy	____
Ulceration	Conservative	Antibiotics	Debridement, surgical management	Hysterectomy	____
Incompetent				Obstetrical management	____

UTERUS / CERVIX

Analytic		
MRI	Assessment of wall thickness, perametrial infiltrates, sinus and fistula formation	Y/N Date:
Ultrasound	Assessment of wall thickness, perametrial infiltrates, sinus and fistula formation	Y/N Date:
EUA cytology / biopsy	Assessment of mucosal surfaces and ulcers	Y/N Date:

OVARY / REPRODUCTIVE

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Hot flashes	Occasional	Intermittent	Persistent		____ **Instructions**
Dysmenorrhea	Occasional	Intermittent	Persistent		____ **Score the 9 SOM parameters with 1 - 4**
Menstruation		Oligomenorrhea	Amenorrhea		____
Objective					
Ovulation			Anovulation in premenopausal women		____ **(Score = 0 if there are no toxicities)**
Involuntary infertility			Infertility		____
Osteoporosis			Radiographic evidence	Fracture	____ **Total the scores and divide by 9**
Management					
Dysmenorrhea, Hot flashes		Hormone replacement			____ **LENT Score:**
Menstruation		Hormone replacement			____
Osteoporosis		Hormone replacement, Calcium supplements			____ _____

Analytic		
FSH/ LH/ Estradiol	Assessment of hormonal production	Y/N Date:
Bone densitometry	Quantify bone density	Y/N Date:

SEXUAL DYSFUNCTION - FEMALE

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					**Instructions**
Dyspareunia	Occasional	Intermittent	Persistent	Refractory	____ **Score the 11 SOM parameters with 1 - 4**
Dryness	Occasional	Intermittent	Persistent	Refractory	____
Desire	Occasional	Intermittent	Seldom	Never	____
Satisfaction	Occasional	Intermittent	Seldom	Never	____
Objective					
Vaginal stenosis / length	> 2/3 normal length	1/3 - 2/3 normal length	< 1/3 normal length	Obliteration	____ **(Score = 0 if there are no toxicities)**
Synechiae			Partial	Complete	____
Frequency		Decreased from normal	Rare	Never	____ **Total the scores and divide by 11**
Orgasm	Occasional	Intermittent	Seldom	Never	____
Management					
Dryness	Hormone replacement	Artificial lubrication			____ **LENT Score:**
Stenosis / Synechiae	Occasional dilation	Intermittent dilation	Persistent dilation	Surgical reconstruction	____ _____
Dyspareunia	Occasional hormone cream	Intermittent hormone cream	Persistent hormone cream		____

Analytic		
Psychosocial	Evaluate Quality of Life / Sexual Satisfaction	Y/N Date:
Vaginal measurement	Evaluate degree of vaginal stenosis	Y/N Date:

MUSCLE / SOFT TISSUE

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					**Instructions**
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____ **Score the 12 SOM parameters with 1 - 4**
Function	Interferes with athletic recreation	Interferes with work	Interferes with daily activity	Complete lack	____
Objective					
Edema	Present / asymptomatic	Symptomatic	Secondary dysfunction	Total dysfunction	____ **(Score = 0 if there are no toxicities)**
Mobility and extremity function	Present / asymptomatic	Symptomatic	Secondary dysfunction	No mobility, frozen	____
Fibrosis	Detectable	≤ 20% of muscle	> 20% - 50% of muscle	> 50% of muscle	____ **Total the scores and divide by 12**
Atrophy	≤10%	> 10% - 20%	> 20% -50%	> 50%	____
Contraction		≤ 10% linear field	>10% - 30% linear field	> 30% linear field	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____ **LENT Score:**
Edema		Compression	Medical intervention	Surgical intervention	____
Mobility and extremity function	Occasional physiotherapy	Intermittent physiotherapy	Persistent physiotherapy or medical intervention	Surgical intervention	____
Fibrosis	Occasional physiotherapy	Intermittent physiotherapy		Surgical intervention	____
Atrophy		Intermittent physiotherapy		Surgical intervention	____

Analytic		
MRI	Development of investigational testing suggested	Y/N Date:

PERIPHERAL NERVES

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Strength		Detectable weakness	Persistent weakness	Paralysis, Transverse myelitis	____
Sensory	Occasional paresthesia, hyperthesia	Intermittent paresthesia	Persistent paresthesia	Paralysis	____
Motor paresis	Occasional	< 50% decrease from base line capabilities	≥ 50% decrease from base line capabilities	Paralysis	____
Objective					
Motor dysfunction	< 20% loss	20% - 30% loss	> 30% - 50% loss	> 50% loss	____
Sensory dysfunction	Paresthesia	Vibration decrease	Decrease to pin prick	Complete anesthesia	____
Reflex	Decreased deep tendon reflex	Absent deep tendon reflex			____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Motor dysfunction			Physical or medical intervention	Surgical intervention	____
Sensory dysfunction			Physical or medical intervention	Surgical intervention	____
Sensory				Neurosurgical intervention	____

Instructions

Score the 11 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 11

LENT Score:

_____

Analytic		
MRI	Assessment of associated muscle atrophy, changes in nerve intensity	Y/N Date:
Nerve conduction studies	Assessment of speed, absence of conduction of electrical impulse	Y/N Date:

GROWING BONE

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Abnormal gait	Slight	Noticeable limp	Severe limp	Unable to walk	____
Disfigurement	Slight, not cosmetically significant	Mild cosmetic deformity	Moderate cosmetic deformity	Severe disfigurement	____
Objective					
Extremities	Mild curvature or length discrepancy < 2 cm	Moderate curvature or length discrepancy 2 cm - 5 cm	Severe curvature or length discrepancy > 5 cm	Epiphysiodesis, severe functional deformity	____
Spine					
Sit/standing height	Mild disproportion	Moderate	Severe		____
Scoliosis	< 5°	5° - 10°	> 10° - 20°	> 20°, interfering with cardiopulmonary function	____
Kyphosis/lordosis	Mild radiographic changes	Moderate accentuation	Severe accentuation		____
Femoral heads	Mild valgus/varus deformity	Moderate valgus/varus deformity	Mild slipped capital femoral epiphysis / epiphyseal widening	Severe slipped capital femoral epiphysis > 60°; avascular necrosis	____
Flat/facial bones	Slight changes, not cosmetically significant	Mild cosmetic deformity	Moderate cosmetic deformity	Profound hypoplasia or functional problem	____
Management					
Extremities		Minimal shoe lift	Moderate shoe lift	Surgical intervention	____
Scoliosis			Brace	Surgical intervention	____
Femoral heads			Pinning	Hip replacement	____
Flat/facial bones				Surgical intervention	____

Instructions
Score the 13 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 13

LENT Score:

_____

Analytic					
Measure growth	No growth retardation	Growth retardation ≤ 1 percentile	Growth retardation > 1 percentile	Growth arrest	Y/N Date:
Radiograph / CT	Assessment of bone integrity				Y/N Date:

MATURE BONE (EXCLUDING MANDIBLE)

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Pain	Occasional & minimal	Intermittent & tolerable	Persistent & intense	Refractory & excruciating	____
Function	Interferes with athletic recreation	Interferes with work	Interferes with daily activity	Complete lack of function	____
Joint movement	Stiffness interfering with athletic recreation	Stiffness interfering with work	Stiffness interfering with daily activity	Complete fixation, necrosis	____
Objective					
Fracture			Partial thickness	Full thickness	____
Mucosa soft tissue			Sequestration		____
Skin over bone	Erythema	Ulcer	Sinus	Fistula	____
Joint movement	< 10% decrease	> 10% - 30% decrease	> 30% - 80% decrease	> 80% decrease	____
Management					
Pain	Occasional non-narcotic	Regular non-narcotic	Regular narcotic	Surgical intervention	____
Function	Occasional physiotherapy	Intermittent physiotherapy	Persistent physiotherapy or medical intervention	Surgical intervention	____
Joint movement	Occasional physiotherapy	Intensive physiotherapy	Corrective surgery		____

Instructions

Score the 10 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 10

LENT Score:

_____

Analytic		
Imaging: density	Assessment for osteosclerosis and osteoporosis	Y/N Date
X-ray	Assessment of bone and joint integrity including linear fracture and displaced fracture	Y/N Date:
Arthography	Assessment of joint integrity	Y/N Date:
Arthroscopy	Evaluation for joint abnormalities	Y/N Date: :

BONE MARROW

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Anemia symptoms		Fatigue	Exhaustion		____
Leukopenia symptoms			Fever		____
Thrombocytopenia symptoms			Easy bruisability	Spontaneous bleeding	____
Objective					
Anemia		Abnormal Hgb / Hct < 10 / < 30	Pallor	Tachypnea	____
Leukopenia		Abnormal WBC < 2000	Infection	Sepsis	____
Thrombocytopenia	Abnormal aspirate / biopsy	Platelets >20K - 100K	Platelets >5K - 20 K, petechia	Platelets < 5K, hemorrhage	____
Management					
Anemia			Occasional use of red blood products	Frequent use of red blood products	____
Leukopenia			Antibiotics / cytokines		____
Thrombocytopenia			Platelets / red blood cells	Bone marrow transplant	____

Instructions

Score the 9 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 9

LENT Score:

_____

Analytic		
Assays	Assessment of bone marrow reserves with:	
	Hematopoietic progenitor cell assays in common use (CFU-GM, BFU-R, CFU-GEMM, CFU-blast. etc./)	Y/N Date:
	Stromal cell assays (CRU-F, support of long-term bone marrow cultures)	Y/N Date:
	Growth factor production	Y/N Date:
	Primitive stem cell assays (HPP-CRL, CFU-Dexter, LTC-IC, somatic mutation analysis / DWA analysis)	Y/N Date:
Chimerism, Clonality	In setting of bone marrow transplant (Studies of mixed (donor/host) chimerism, studies of clonality (donor vs host))	Y/N Date:
	Future consideration: challenge with growth factors to assay stem cell reserve	

SKIN / SUBCUTANEOUS TISSUE

	GRADE 1	GRADE 2	GRADE 3	GRADE 4	SCORING
Subjective					
Scaliness/Roughness	Present / asymptomatic	Symptomatic	Require constant attention		____
Sensation	Hypersensitivity, pruritus	Intermittent pain	Persistent pain	Debilitating dysfunction	____
Objective					
Edema	Present / asymptomatic	Symptomatic	Secondary dysfunction	Total dysfunction	____
Alopecia (scalp)	Thinning	Patchy, permanent	Complete, permanent		____
Pigmentation change	Transitory, slight	Permanent, marked			____
Ulcer / Necrosis	Epidermal only	Dermal	Subcutaneous	Bone exposed	____
Telangiectasia	Minor	Moderate < 50%	Gross ≥ 50%		____
Fibrosis / Scar	Present / asymptomatic	Symptomatic	Secondary dysfunction	Total dysfunction	____
Atrophy / Contraction (depression)	Present / asymptomatic	Symptomatic / < 10%	Secondary dysfunction / 10% - 30%	Total dysfunction / > 30%	____
Management					
Dryness			Medical intervention		____
Sensation		Intermittent medical intervention	Continuous medical intervention		____
Ulcer			Medical intervention	Surgical intervention/ amputation	____
Edema			Medical intervention	Surgical intervention/ amputation	____
Fibrosis / Scar			Medical intervention	Surgical intervention/ amputation	____

Instructions
Score the 14 SOM parameters with 1 - 4

(Score = 0 if there are no toxicities)

Total the scores and divide by 14

LENT Score:

_____

Analytic Color photographs	Assessment of changes in appearance	Y/N Date: